

HENRY DIBIE

Backend Engineer (Node.js / TypeScript)

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SUMMARY

Backend engineer with 5+ years of industry experience across fintech, telecom, and economic research, and production delivery in Node.js, TypeScript, and Python. Owns full-scope backend engineering for a live multi-vendor marketplace platform built on Node.js, TypeScript, and Firebase Cloud Functions, across five delivery phases and 390 acceptance tests, including a provider-agnostic subscription engine spanning three payment providers and a rule-based moderation system with trust scoring. Maintains a published open-source Node.js/TypeScript developer tool (Schema-Watch, npm) with its own CI pipeline. Also brings production data engineering and ML depth (dbt, Prefect, PySpark, FastAPI, XGBoost) from prior fintech and research roles.

TECHNICAL SKILLS

- **Languages & Runtimes:** TypeScript, JavaScript (Node.js), Python, SQL
- **Backend & APIs:** Firebase Cloud Functions, Firestore security rules, REST API design, FastAPI, PostgreSQL, SQLAlchemy
- **Infrastructure:** Firebase, Azure, AWS (ECS Fargate), GCP, Docker, GitHub Actions (CI/CD)
- **Data & Automation:** Prefect, dbt, PySpark, Playwright, BeautifulSoup, HubSpot API, Fathom API, Slack API, Google Docs API
- **Testing & Quality:** Automated acceptance testing against the Firebase Emulator Suite, pytest

FEATURED BACKEND PROJECTS

Multi-Vendor Marketplace Platform: Backend & Web | Freelance

2025 – Present

Node.js, TypeScript, Firebase Cloud Functions, Firestore, Next.js | github.com/HenryMorganDibie/multivendor-marketplace-platform

- Designed and built the backend across five delivery phases (identity/auth, catalog/orders/inventory, real-time chat/notifications/support, vendor subscriptions/ratings/invoices, and a landing page/CMS/vendor portal package), covering both a customer-facing mobile app and a Next.js web frontend.
- Wrote 390 acceptance tests across all five phases, run against the Firebase Emulator Suite, including denial-path as well as success-path coverage for every security rule.
- Architected a provider-agnostic subscription engine live on Paystack, Flutterwave, and Stripe: a single shared webhook state machine handles idempotency, staleness rejection, distributed locking, and out-of-order event resolution, with each provider file responsible only for its own signature verification and payload normalization.
- Built a server-side plan-gating matrix enforced across catalog limits, order automation, and vendor features, with Firestore as the single source of truth and no application code ever reading a payment provider's API directly.
- Built a rule-based chat and catalog moderation engine using cumulative per-user trust scoring with transactional escalation, avoiding false positives on ordinary commerce language while still catching genuine policy violations.
- Built a ratings system with strict vendor-side anonymity (vendors never receive order or customer identifiers) enforced at the security-rules layer, not just by convention, and an invoicing system with server-computed totals, plan-gated branding, and PDF generation via pdfkit.
- Built the Next.js frontend consuming this backend: a 14-page public landing site, a CMS editor for draft and publish versioning, and a vendor portal, 26 routes typechecked and production-build-verified.
- Replaced per-provider checkout with a single provider-neutral flow: one public callable resolves the vendor's country server-side, selects a payment provider from a private priority list, and prevents double-billing on plan upgrades by cancelling the existing subscription before starting a new one. Also built the invoice/receipt renumbering scheme and the price-change policy protecting existing subscribers from a live pricing change until their next real renewal.
- Documented an honest, specific security posture for the codebase, including what has been verified through self-review and testing versus what remains open ahead of an independent audit.
- Audited a disconnected admin console frontend against the live backend: reconciled two conflicting backend-contract layers in the client codebase, corrected an initial 76-endpoint estimate down to a verified 51 through direct source review, and produced a phased implementation plan covering a second, independent RBAC permission system, structural audit logging across every mutation, and platform-wide analytics aggregation with no prior equivalent in the codebase.

Schema-Watch: Open-Source API Contract Diffing Tool | Personal / Open Source

2025 – Present

TypeScript, Node.js, npm | github.com/HenryMorganDibie/schema-watch

- Built and published an npm package (npx schema-watch) that diffs live API response shape against a committed baseline, flags breaking changes, identifies affected frontend files, and fails CI builds.
- Architected as a TypeScript monorepo: a core diff engine, a local reverse-proxy agent with SQLite and REST/WS APIs, a live dashboard, and a separate hosted Cloud backend for team-wide contract monitoring, billing, and Slack/Discord alerting.
- Shipped a GitHub Actions integration that comments on pull requests, publishes a required check run, and uploads SARIF findings; the tool runs against itself in its own CI on every pull request.

New Care Health Partners: Compliance Portal Backend & AWS Architecture | Freelance 2025 – Present

- Designing backend and AWS architecture for a healthcare compliance portal: database design, REST API design, role-based authorization for admin, applicant, and employee user types, secure private document storage, automated expiry and reminder jobs, audit logging, and backend security hardening ahead of an independent penetration test.

Chirp API: Production Audit & Stabilization | Freelance 2025 – Present

Python 3.12, gRPC, SQLAlchemy 2.0 | github.com/HenryMorganDibie/chirp-api-refactoring

- Audited an unfamiliar production-grade gRPC monorepo and found passwords hashed with SHA-256 and a static salt, plus a hardcoded JWT secret fallback that would let anyone with code access forge session tokens for any user, including admins.
- Fixed both: migrated to bcrypt (cost factor 12) with a zero-downtime path that detects and rehashes legacy passwords on login, no forced reset; made the JWT secret fail server startup if unset rather than falling back to a hardcoded value. Covered by 15 new tests.
- Found and fixed an N+1 query pattern across every list endpoint, reducing a 10-post home feed from 31 queries to 3 (about 90%) via a single bulk aggregation query, with 6 regression tests that count actual DB round trips so a future regression fails CI automatically.
- Added UUID trace IDs per request and a unified error-classification layer mapping exceptions to correct gRPC status codes; found and fixed real CI/monorepo config bugs, including a missing lockfile in task inputs that could let a stale environment pass CI silently.

EXPERIENCE

Freelance ML Systems Engineer & Data Scientist | Self-employed Nov 2025 – Present

- Own full-scope backend engineering for the Multi-Vendor Marketplace Platform and backend/AWS architecture for New Care Health Partners (see Featured Projects above).
- Built and published Schema-Watch, an open-source Node.js/TypeScript API contract-diffing tool (see Featured Projects above).
- Delivered the MVNO Intelligence Hub for a telecom client (Azure, FastAPI, Prophet forecasting, 30K subscribers, 5.4M records load-tested) and the knowledge-rag-api for a client (pgvector, BM25, reciprocal rank fusion, cross-encoder reranker, AWS ECS Fargate).
- Built The Scribe, an AI writing assistant: FastAPI backend, async SQLAlchemy/PostgreSQL with pgvector, JWT auth, and Dramatiq/Redis background workers. SSE-streamed generation grounded in a RAG-retrieved voice profile per author, an LLM provider abstraction (Anthropic/Groq), and a full smoke-test suite covering the user journey end to end.
- Reviewed a GPU inference observability platform for a client exploring product-market fit (inter-token latency, time-to-first-token on vLLM workloads).

Data Research Analyst, Automation & AI | Stears 2024 – 2025

- Built and maintained data pipelines using dbt, Prefect, and PySpark to process and model economic and market data across 54 African markets.
- Built and scheduled production data pipelines (Python, Node.js) extracting company profile data, including executive and non-executive leadership, using Playwright and BeautifulSoup for web scraping, integrated with the Google Docs and Slack APIs for review and distribution.
- Built a GenAI sales intelligence pipeline that pulled customer sales calls from HubSpot and Fathom via their APIs, then ran Whisper transcription, OpenAI embeddings, and FAISS vector search to surface actionable insights from sales conversations.
- Owned automation tooling end to end, from data ingestion through transformation to delivery, and reviewed analyst-submitted data before publication to the company website.

Credit Risk Analyst | Renmoney Microfinance Bank 2022 – 2024

- Built probability-of-default models using XGBoost and logistic regression, applied to a regulated lending portfolio, improving approval precision by 15%.
- Used SHAP for model interpretability, translating model outputs into explainable risk drivers for credit policy stakeholders.
- Performed bureau data analysis using CRC and FirstCentral credit bureau data to inform underwriting decisions.
- Designed and ran A/B and champion-challenger tests to validate model performance before production rollout.
- Built Power BI dashboards for ongoing portfolio risk monitoring and reporting.

Data Scientist | Nupat Technologies 2020 – 2022

- Consulted to client companies on insurance claims data science (models and analysis pipelines for claims processing) and supported internal data science work for customers learning through Nupat's code camp.

EDUCATION, CERTIFICATIONS & RECOGNITION

- Professional Certification in Data Analytics, Turing College (Digital Explorers II Scholar, 2024)
- BSc, Business Administration, École Supérieure de Gestion et de Technologie
- 1st Place, Splendor Analytics Trial Activation Challenge (94/100, Distinction)